



CH₃SH conversion in a tubular flow reactor. Experiments and kinetic modelling



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ABSTRACT

The use of non-conventional fuel sources, such as shale gas, brings new research requisites for its proper use in an environmental friendly manner. In this context, shale gas may include different sulphur containing compounds, such as methanethiol, that is also formed as intermediate during sulphur containing residues processing. The present work includes an experimental and kinetic modelling study of the oxidation of methanethiol, CH₃SH, in a quartz flow tubular reactor at atmospheric pressure and in the 300–1400 K temperature range. The influence of the temperature, the O₂ concentration and the presence of H₂O on the conversion regime of CH₃SH and the formation of different compounds has been analysed. The experimental results have been interpreted in terms of a detailed gas-phase mechanism compiled in the present work, and the elementary steps involved in the conversion of CH₃SH have been identified. In general, oxidation of CH₃SH is favoured by both oxygen level and temperature, while the presence of H₂O does not modify the CH₃SH conversion profile. The main sulphur containing products are SO₂, H₂S and CS₂, pointing to a significant role of other products, apart from SO₂, for the control of pollutant emissions.

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1. Introduction

The use of non-conventional fuel sources, such as shale gas, refinery gas, or biogas obtained from anaerobic digestion, brings new research requisites for its proper use in an environmental friendly manner. In this context, these gases, apart from CH₄ and CO₂, may include different sulphur containing compounds, such as hydrogen sulphide and mercaptans as methanethiol, that is also formed as intermediate during sulphur containing residues processing. The presence of sulphur in sour gases makes more difficult its application, mainly because of environmental issues. While the traditional way to use these fuels is based on their exhaustive cleaning [e.g. [1]], the present strategies include the direct use of sour gases in combustion together with the improvement of technologies and combustion processes, along with the later solution of environmental problems. This is what has been named as a “double revolution” [2].

Methanethiol, CH₃SH, also known as methyl mercaptan, is a sulphur organic compound that can be present in a number of fuel gases, such as natural gas, sour gas, shale gas, biogas or refinery gas [3]. Formation of mercaptans (mainly CH₃SH), together with COS and CS₂ also occurs in the Claus process through the

interaction of CO₂, hydrocarbon radicals and sulphur species [4], reducing thus the sulphur recovery efficiency. Shale gas has been reported to contain typically a total sulphur content of up to 1%, including sulphur in pyritic form, sulfates and organics [5], with concentrations of mercaptans (CH₃SH) of up to 0.5% [6]. CH₃SH can also be formed as an intermediate compound in the destruction of mustard gas, ClCH₂–CH₂–S–CH₂–CH₂Cl, through incineration mainly chosen as a high effectiveness destruction method, even though there is still significant concern over the potential formation of toxic emissions [7,8]. Under oxidation conditions, conversion of CH₃SH may lead preferably to SO₂, but also to H₂S or other carbon-sulphur compounds such as COS or even CS₂.

While the conversion of COS and CS₂ have been considered in the past [e.g. [9–11]], to our knowledge, no investigation of the oxidation of CH₃SH has been carried out under combustion conditions. However, significant efforts have been drawn to evaluate the oxidation of CH₃SH in the atmosphere, because of its probable and significant role in the atmospheric sulphur cycle (e.g. [12–14]). Actually, methanethiol has been reported to represent around a 10% of the global flux of sulphur compounds in the atmosphere [15].

In this context, the present study aims to carry out a combined experimental and kinetic modelling study of the conversion of CH₃SH under combustion conditions. The impact of temperature and stoichiometry, ranging from almost pyrolysis to very oxidizing conditions, on the process is evaluated in a tubular flow reactor

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at atmospheric pressure and in the 300–1400 K temperature range. Experimental results are discussed in terms of a detailed kinetic mechanism. The present study is of interest in its own right, and may serve as a base case to consider the behavior of non-aromatic organic sulphur in combustion systems.

2. Experimental methodology

Most of the experiments have been carried out in a quartz tubular flow reactor at atmospheric pressure. Only a brief experimental setup (setup 1) description is given here and a more detailed description can be found elsewhere [16]. The reactor has an isothermal reaction zone of 20 cm in length and 0.87 cm of internal diameter. Total flow rate in all experiments was 1 L (STP)/min, resulting in a gas residence time as a function of temperature of $194.6/T(K)$, in seconds. The reactor is placed in a three-zone electrically heated oven, ensuring a uniform temperature profile (± 5 K) along the reaction zone. Experiments are carried out under highly diluted conditions using nitrogen as bath gas. Gases from cylinders are led to the reactor in up to four separate streams, following the procedure of Alzueta et al. [17], and are heated separately and mixed in cross flow at the reactor inlet. At the outlet of the reaction zone, the product gas is quenched by means of external cooling air. The exhaust gases go through a condenser and a filter, removing any possible residual solid and moisture, and subsequently conducted to the analysis system, which includes a UV continuous analyser for SO_2 , an IR continuous analyser for CO and CO_2 and a gas micro-chromatograph for CH_3SH , H_2S , COS, CS_2 , CH_4 , CO, CO_2 , O_2 and H_2 quantification. The uncertainty of the measurements is estimated within 5%.

Additionally, with the purpose of evaluating a strange behavior found in CO formation that will be described later in detail, one experiment has been performed in a different installation (setup 2) that is described in a previous work by our group [18].

3. Kinetic model

The experimental results have been interpreted in terms of an updated kinetic model. The mechanism used is based on the mechanism by Alzueta et al. [19] developed to analyse the impact of the presence of SO_2 on fuel conversion, and includes the interaction of SO_2 with CO and small hydrocarbons by Glarborg et al. [20,21]. This mechanism was later updated by Giménez-López et al. [22] to account for oxy-fuel combustion conditions. To the updated mechanism, we have added a CH_3SH conversion subset, and related thermochemistry, taken mainly from the works of Zheng et al. [7,23] and Van de Vijver et al. [24], who studied respectively the oxidation of diethyl sulphide and the pyrolysis of alkyl sulphides. The work of Van de Vijver et al. [24] is, in turn, largely based on previous works by the group [25–29]. This reaction subset has also been largely used in the recent work of Gersen et al. [30], who studied the effect of H_2S addition on methane ignition and oxidation at high pressure. Subsets for COS and CS_2 conversion were taken from previous works from the authors [31,32] and the H_2S conversion subset from a recent work by our group [33]. This H_2S reaction subset is largely based on the works of Sendt et al. [34], Zhou et al. [35] and Song et al. [36]. The mechanism of Colom-Díaz et al. [33] included the isomerization of HSO_2 to HSO_2 , as a key fast step for H_2S oxidation, together with new kinetic parameters for the $\text{SH} + \text{H}_2\text{O}_2 \rightleftharpoons \text{H}_2\text{S} + \text{HO}_2$ reaction, which was found to have a significant impact on H_2S conversion, and that is included in the present mechanism, since the formation of significant amounts of H_2S from methanethiol can be expected.

The final reaction mechanism listing is included as supplementary material. As for thermochemical data, same sources as for the

Table 1

Experimental conditions. Experiments 1 to 7 are performed at atmospheric pressure in setup 1 [16]. Experiment 8 is performed at 1.81 bar in setup 2 [18].

Set	CH_3SH (ppm)	O_2 (ppm)	H_2O (%)	λ	Setup
1	983	42	–	0.01	1
2	968	1211	–	0.42	1
2a	972	979	0.5	0.34	1
3	977	2269	–	0.77	1
4	952	2841	–	0.99	1
4a	1003	2953	0.5	0.98	1
4r	965	2841	–	0.98	1
5	965	4576	–	1.58	1
6	973	5878	–	2.01	1
7	973	14,855	–	5.08	1
7a	1015	13,370	0.5	4.39	1
8	987	3025	–	1.02	2

corresponding reactions were used. Calculations were carried out using the PFR model of the Chemkin Pro suite [37].

4. Results and discussion

The study of CH_3SH oxidation in a tubular flow reactor at atmospheric pressure has been carried out from almost pyrolysis to fuel-lean conditions, in the temperature range of 300–1400 K. The experimental conditions studied are listed in Table 1. The influence of the amount of oxygen available on the process was studied for different values of λ , defined as $\text{O}_2(\text{real})/\text{O}_2(\text{stoichiometric})$. For an inlet total flow rate of 1 L (STP)/min, the gas residence time in the reactor varies in the 0.14–0.24 s range depending on temperature, according to $194.6/T(K)$, with the exception of experiment 8, for which the residence time is $414/T(K)$ seconds.

For the conditions listed in Table 1, the concentrations of CH_3SH , SO_2 , H_2S , O_2 , CO, CO_2 , H_2 , CH_4 and CS_2 obtained as a function of temperature are presented in Fig. 1.

Figure 1 shows the experimental results for the quantified species during the conversion of CH_3SH as a function of temperature and for the different stoichiometries studied (λ values ranging from almost pyrolysis conditions to very fuel-lean conditions), in the absence of water. Experimental sulphur and carbon mass balances close to $100 \pm 5\%$ in most cases, except for carbon at intermediate temperatures (823–923 K) where the balance goes down to 70–75%. This is attributed to the fact that the sharpest conversion of CH_3SH occurs in this temperature interval, with many intermediates in small concentrations that appear and are not quantified.

Methanethiol starts its conversion between 800 and 1000 K as a function of stoichiometry, with the onset conversion temperature shifted towards lower temperatures as the oxygen availability in the reaction environment is higher. Most of CH_3SH conversion occurs in a narrow temperature range of about 50 K. The final products dominant in CH_3SH conversion include H_2 , CO, CO_2 , and SO_2 as sulphur species. With the exception of CO and SO_2 (under fuel-rich conditions), which exhibit a maximum as a function of temperature, the rest of dominant final products increase with temperature under the conditions considered. It is worthwhile to mention that, for the highest temperatures (above 1000 K approximately) and the fuel leaner stoichiometries ($\lambda = 1.5$ and above), the sulphur present in CH_3SH is fully converted to SO_2 . For $\lambda < 1$, conversion of CH_3SH also results in the formation of intermediates, as CH_4 and H_2S , which exhibit a maximum that is shifted to lower temperatures, as the stoichiometry becomes fuel leaner. It is noticeable that, under the fuel-rich conditions studied, CH_4 and H_2S are still present even at the highest temperatures considered. It is also remarkable that CS_2 is formed in all the experiments, appearing in significant concentrations under fuel-rich conditions, while COS ($\text{O}=\text{C}=\text{S}$) was found in the range of few ppm, only under reducing conditions ($\lambda < 0.7$) and at the highest

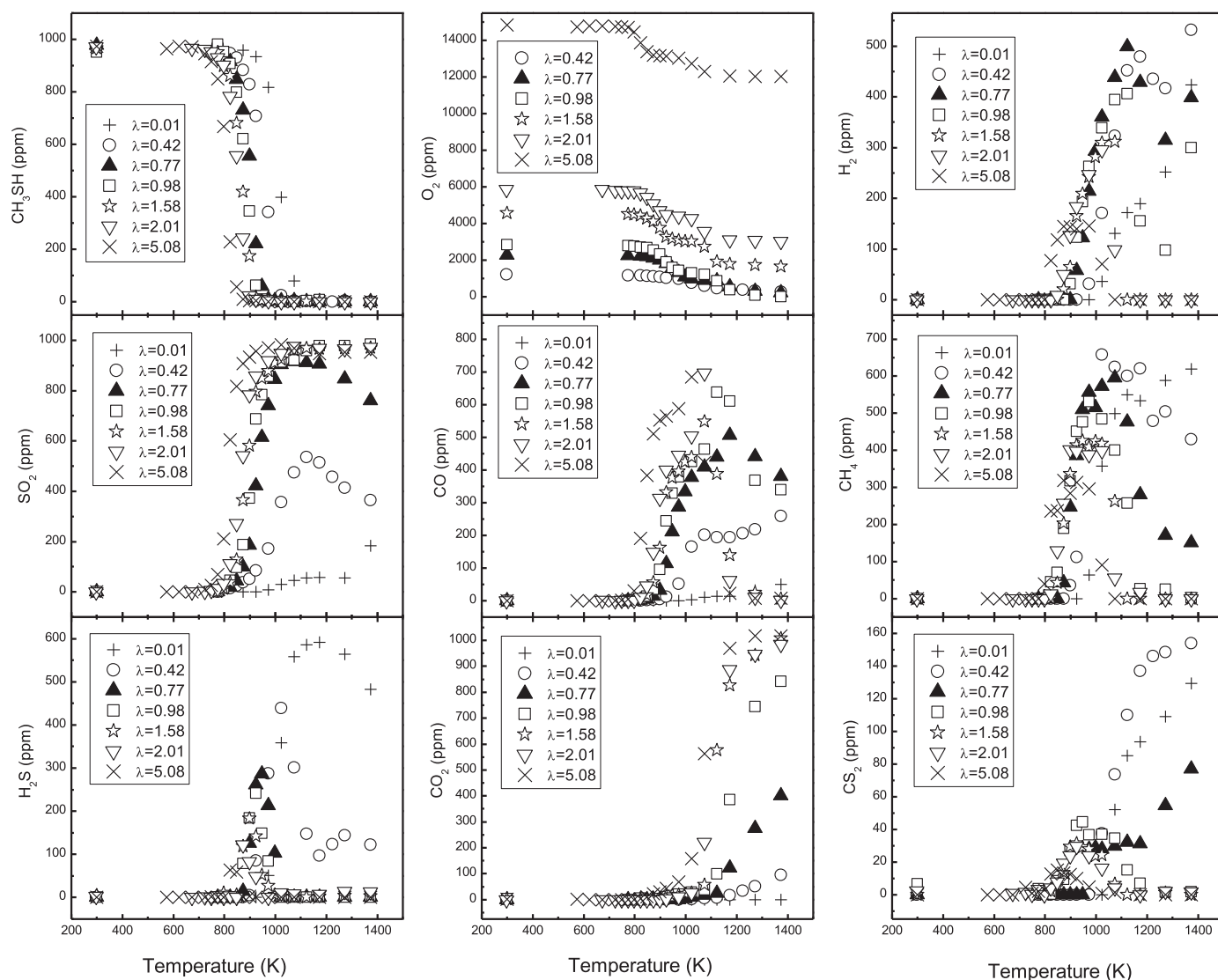


Fig. 1. Experimental concentrations of CH_3SH , SO_2 , H_2S , O_2 , CO , CO_2 , H_2 , CH_4 and CS_2 as a function of temperature for different stoichiometries. Results correspond to sets 1 to 7 in Table 1.

temperature considered in the present study. This is in agreement with earlier studies [31], where the formation of COS has been reported to be formed in the oxidation of CS_2 . Under reducing conditions, the formation of a yellow powder was observed at the outlet of the reactor, but its quantification was not possible, as also happened in previous works by our group [33]. The kinetic model used in the present work also predicts the formation of small amounts of S_2 under reducing conditions.

The results shown in Fig. 1 correspond to experiments performed in the absence of water vapour. In order to evaluate the impact of the presence of water, which has been reported to shift the conversion of CO in the presence of SO_2 [20], experiments in the presence of water (0.5%) have been performed, and the results obtained in both the absence and presence of water are shown in Fig. 2. As an example, for fuel-rich conditions, Fig. 2a shows that the results with and without water are very similar for all the species measured, despite the slight differences between the two experiments in relation to the initial concentrations of CH_3SH and O_2 (968 and 972 ppm, and 1211 and 979 ppm, respectively), which result in a slightly different λ value (0.42 and 0.34 respectively), and the slightly different pressures (1 and 1.8 bar respectively).

Similar findings were obtained for other stoichiometries (Fig. 2b and c). From these results, and under the conditions considered in the present work, no impact of water presence on the conversion of CH_3SH can be drawn.

In order to further explore the performance of the experiments, Fig. 3 shows an example of the repeatability of experiments for the case of $\lambda = 1$. The results show a good repeatability of all the different species concentrations measured, and the slight differences observed can be attributed to the non-exactly equal initial conditions of experiments 4 and 4r in Table 1.

As has been mentioned earlier, we have used a chemical detailed mechanism to interpret the experimental data of methanethiol conversion under the studied conditions in the present work. Figure 4 shows, as an example, the comparison between the experimental results and calculations for three selected stoichiometries (fuel-rich: $\lambda = 0.77$, stoichiometric: $\lambda = 0.99$, and fuel-lean: $\lambda = 5.08$, sets 3, 4 and 7 in Table 1). Symbols represent experimental data and lines indicate model predictions using the mechanism compiled in this work. As seen, the model reproduces fairly well both the tendencies and the concentrations of the different compounds, through all the temperature range considered

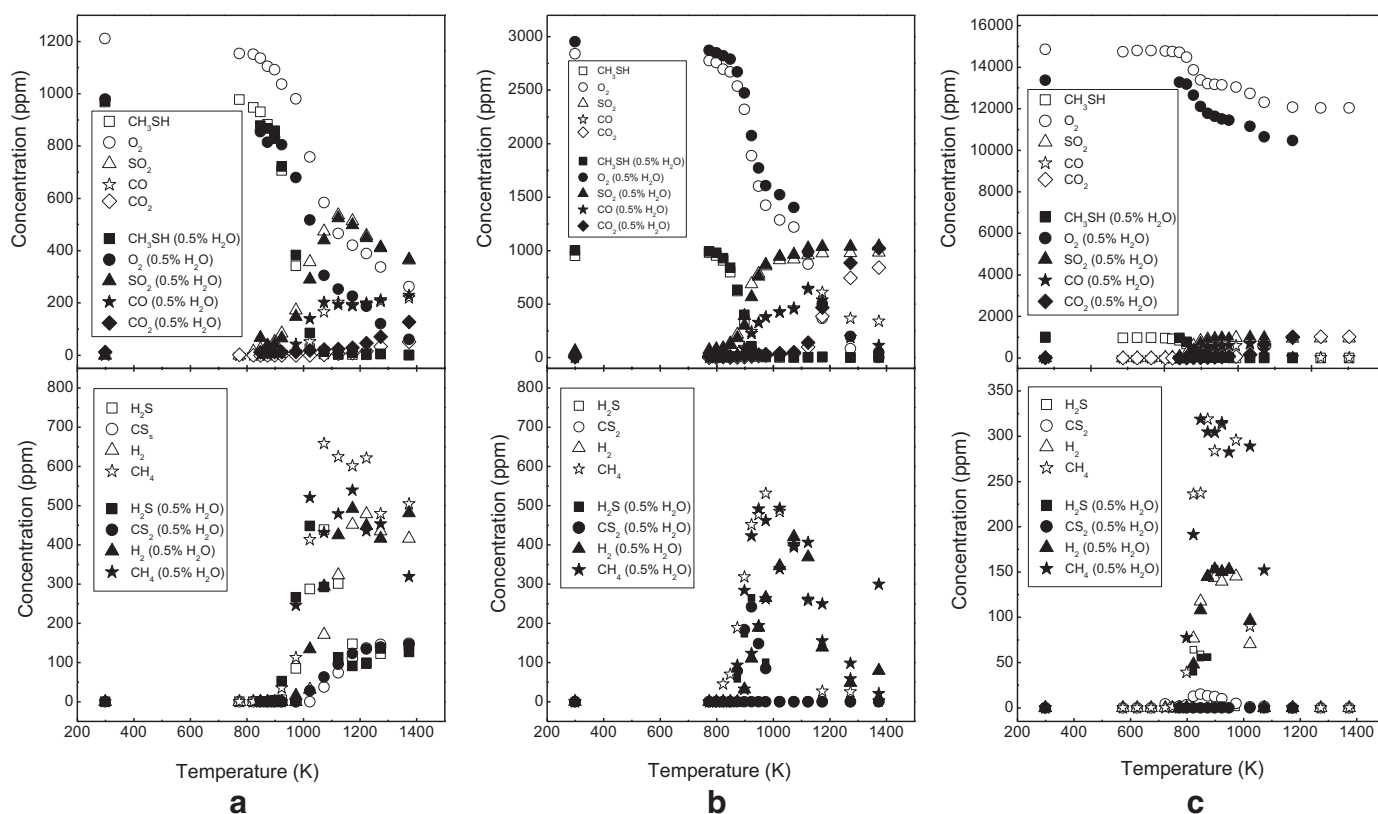


Fig. 2. Experimental concentrations of CH₃SH, SO₂, H₂S, O₂, CO, CO₂, H₂, CH₄ and CS₂ as a function of temperature both in the absence and presence of water (0.5%) and for rich, stoichiometric and fuel-lean conditions. Results of (a) graphs correspond to sets 2 and 2a (fuel-rich conditions); results of (b) graphs correspond to sets 4 and 4a (stoichiometric conditions); and results of (c) graphs correspond to sets 7 and 7a (fuel-lean conditions) in Table 1.

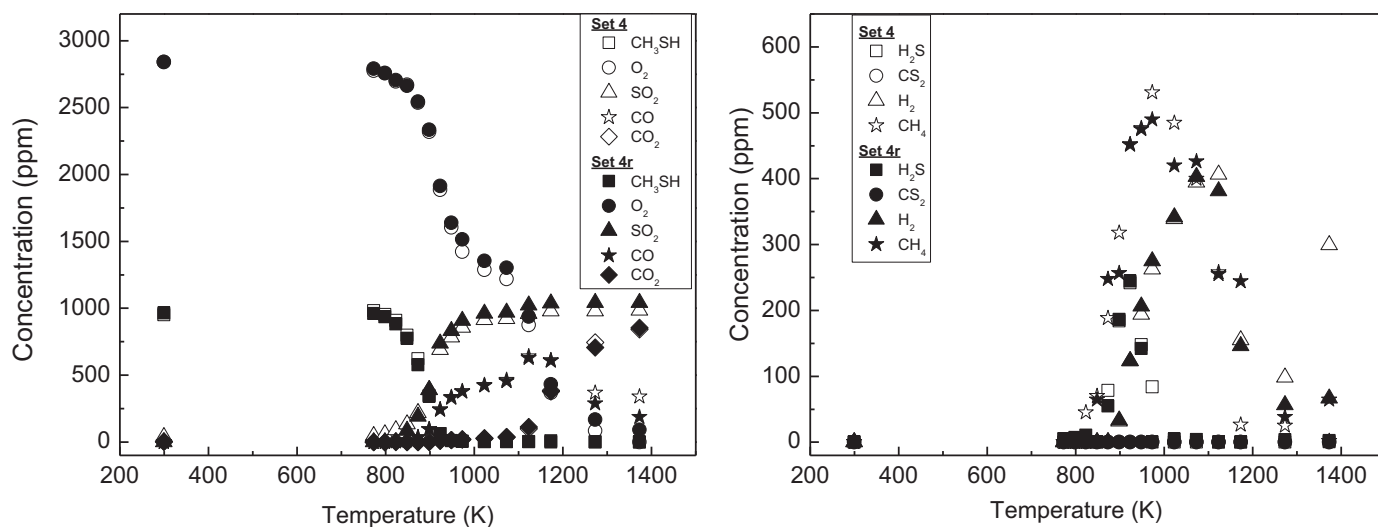


Fig. 3. Example of the repeatability of experiments. Experimental data correspond to experiments 4 and 4r in Table 1.

and for the different stoichiometries, with the exception of CO at temperatures below approximately 1200 K. A similar agreement is obtained for the rest of stoichiometries studies, not shown.

While the experimental CO concentration exhibits an early formation coinciding with CH₃SH diminution, the model does not capture this behaviour. It has to be mentioned that the concentration of CO in the experiments was measured simultaneously by a continuous IR analyser and by gas chromatography, and identical results were obtained. Additionally, we also performed some selected FTIR measurements and the FTIR CO determinations

coincided with the CG and continuous analysers. Therefore, and in order to further prove if this early experimental CO formation is an experimental artefact or a phenomenon effectively occurring during CH₃SH conversion, we decided to do an additional experiment in a different experimental setup [18], which has been used with success in other studies addressing sulphur chemistry (e.g. [33]). This setup, which has significantly different dimensions compared to the one used in the rest of experiments, includes a flow tubular reactor where reactants enter premixed. The results of this experiment (experiment 8 in Table 1), together with model

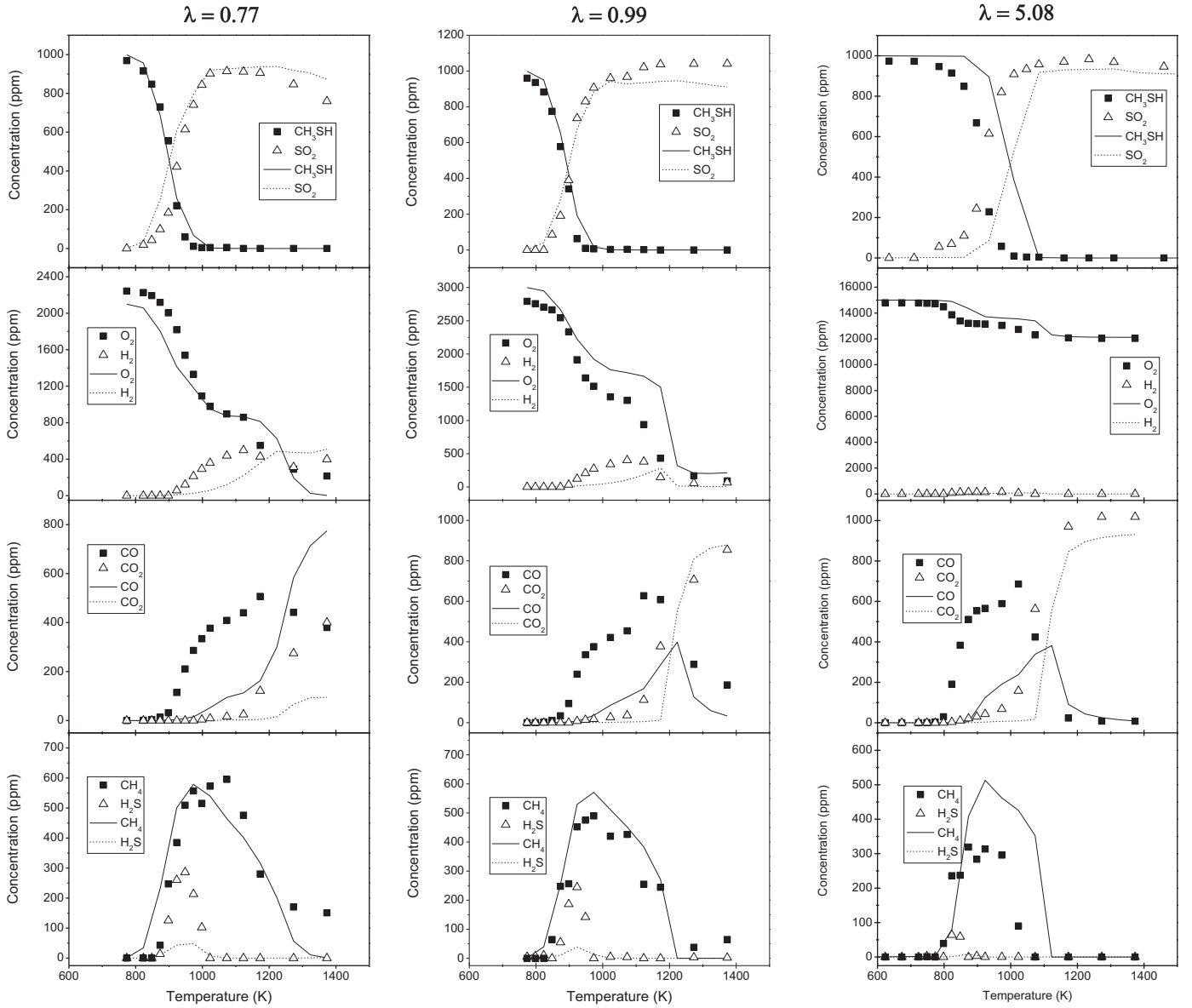
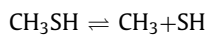
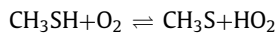
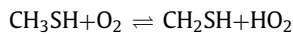


Fig. 4. Comparison between simulations and experimental results for $\lambda = 0.77$ (set 3 in Table 1), $\lambda = 0.99$ (set 4 in Table 1), and $\lambda = 5.08$ (set 7 in Table 1).

calculations, are shown in Fig. 5. It can be observed that, despite the differences in the experimental conditions and installation used, the model can reproduce the experimental observations with the exception of the concentration of CO in the 800 to 1200 K temperature range, where the same early formation of CO is attained. Thus, from these results, we can conclude that the early formation of CO effectively occurs. However, we are not able to predict this early CO formation with the current knowledge of the chemistry addressing the conversion of CH_3SH and other thiols.

Figure 6 includes a reaction pathway diagram of CH_3SH conversion. At the beginning of reaction, CH_3SH is mainly decomposed or consumed by reaction with O_2 (R1 and R2), with a minor contribution of thermal decomposition (R3),



SH radicals contribute to S and S_2 formation, which appear to be relevant for CS_2 formation, as stated further below. A small fraction of SH radicals reacts with oxygen and follows the typical oxidation pathway for this species, as shown in Colom-Díaz et al. [33]. The SH radicals formed in reaction R3 also participate in a minority pathway, reacting with C_2H_4 , which in turn is formed from ethane generated from CH_3 recombination, leading to $\text{CH}_2\text{CH}_2\text{SH}$, which participates in an isomerization cycle to $\text{CH}_3\text{CH}_2\text{S}$ and goes back to $\text{CH}_2\text{CH}_2\text{SH}$.

Apart from the $\text{CH}_3\text{SH} + \text{O}_2$ reaction, as soon as methyl radicals are formed (e.g. reaction R3), they react with CH_3SH leading to the formation of two isomers, CH_3S and CH_2SH , (reactions R4 and R5):



and once the O/H radical pool builds up, reactions with radicals also occur (R6–R9), with reaction (R6) as the dominant one:

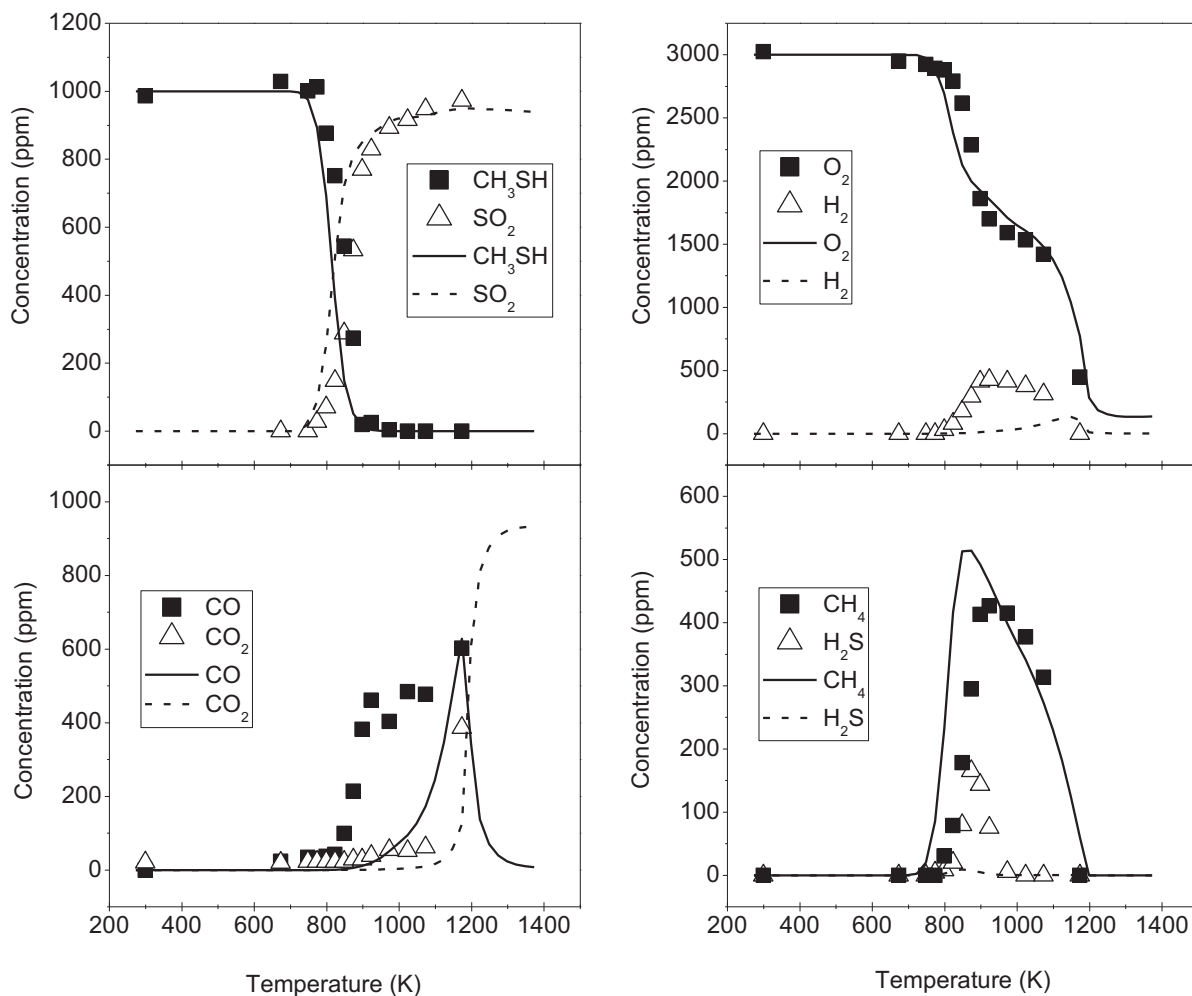


Fig. 5. Experimental and simulated results of experiment 8 in Table 1.

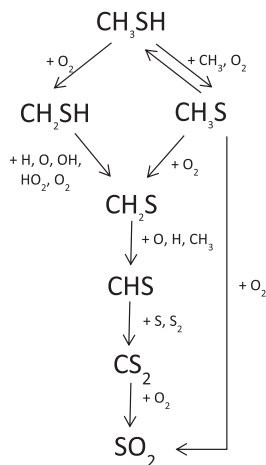


Fig. 6. Main reaction pathway diagram of CH_3SH conversion to SO_2 .

The CH_3SH subset was taken entirely from the work of Van der Vijver et al. [24], which was developed for the thermal decomposition of diethyl sulphide and ethyl methyl sulphide using a rule based kinetic model generator Genesys, with certain modifications as discussed later, and from the works of Zheng et al. [7,23] of their studies on ethyl methyl sulphide and diethyl sulphide oxidation. As seen, the mechanism describes properly methanethiol conversion and, thus, it can be considered as appropriate.

The methyl thiyl radical (CH_3S), which is formed in considerable amounts mainly through reaction R4, basically reacts with O_2 , independently of the stoichiometry. The $\text{CH}_3\text{S}+\text{O}_2$ reaction has three product channels (R10–R12):



with the first two ones dominant under the conditions of the present work.

For this reaction ($\text{CH}_3\text{S}+\text{O}_2 \rightarrow \text{products}$), there are only experimental room temperature upper-limit determinations [38,39]. More recently, the reaction was studied by Zhu and Bozzelli [40], who analysed a number of possible reaction channels for this reaction relevant for atmospheric chemistry by means of DFT and ab initio methods, and isolated the dominant channels which



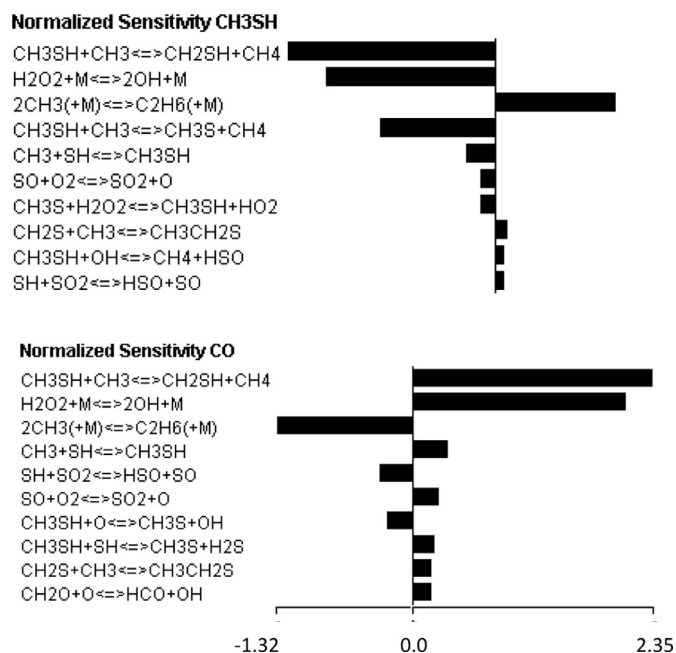


Fig. 7. Normalized sensitivity analysis results for CH₃SH and CO, for $\lambda = 0.99$ (set 4 in Table 1) at the outlet of the reaction zone at the temperature of 798 K.

appear to be those mentioned above. In the present work, the determinations of Zhu and Bozzelli [40] for this reaction rate have been adopted.

CH₂SH, formed in reaction R1, is mainly converted into the CH₂S radical by reaction with oxygen,



for which the rate constant provided by Zheng et al. [23] has been used.

CH₂S is an important intermediate, originated from both CH₃S and CH₂SH (reactions R10 and R13, respectively). Once formed, CH₂S reacts with the O/H radical pool and also with CH₃ radicals, which are present in significant amounts for all the conditions studied, through reactions R14–R17, giving CHS.



CS₂ is formed through the reaction of CHs with S or S₂, formed from SO and SH, in reactions R18 and R19:



CS₂ is finally directly converted into SO₂. Compared to the CH₃S+O₂ $\rightleftharpoons$ CH₃+SO₂ reaction path forming SO₂, the CH₃SH $\rightarrow$ CH₂SH/CH₃S $\rightarrow$ CH₂S $\rightarrow$ CHS $\rightarrow$ CS₂ $\rightarrow$ SO₂ sequence is more relevant as the stoichiometry becomes fuel richer, being dominant for the richest conditions studied in the present work.

Figure 7 shows, as an example, the sensitivity analysis performed for CH₃SH, at $\lambda=0.99$, at the temperature of 798 K and at the outlet of the reaction zone. As seen, the results are mainly

sensitive to the CH₃SH+CH₃ reaction producing CH₂SH, which is an important conversion pathway for CH₃SH once conversion has been initiated, and the dominant one under rich conditions. Reactions of CH₃SH+CH₃ have been taken from the mechanism of Gersen et al. [30] and were also included in the mechanism of Zheng et al. [23], used for the study of diethyl sulphide oxidation with success and to evaluate the pyrolysis and oxidation of sulphur mustard simulates [41]. The kinetic parameters of the CH₃SH+CH₃ reaction are theoretical estimates. Therefore, since these reactions appear to show a significant sensitivity under the studied conditions of this work, a proper determination of them, not available at present, would be desirable. Also, reactions controlling the radical H/O and CH₃ pool appear to exhibit a high sensitivity on the model calculations for both CH₃SH and CO, which are representative of the full oxidation process of methanethiol. Similar results, not shown, are obtained for other stoichiometries considered.

5. Conclusions

Oxidation of CH₃SH at atmospheric pressure has been studied under different reaction atmospheres, varying the air excess ratio (λ) from almost pyrolytic conditions ($\lambda=0.01$) to oxidizing conditions ($\lambda=5.08$). The experiments were carried out in a quartz tubular flow reactor, in the 300–1400 K temperature range and concentrations of CH₃SH, H₂S, SO₂, COS, CS₂, CH₄, H₂, O₂, CO and CO₂ were determined. CH₃SH conversion starts between 800 and 1000 K, shifting toward lower temperatures as λ is higher, and most of the conversion occurs in a narrow temperature window. The results obtained with and without water vapour are similar for all the species measured. The main products of CH₃SH conversion appear to be H₂, CO, CO₂ and SO₂, with a remarkable formation of CS₂, while COS only was detected under reducing conditions ($\lambda<0.7$) and at the highest temperatures studied.

A detailed kinetic mechanism for the conversion of CH₃SH under the present conditions has been compiled based on different literature reaction subsets, and has been used to simulate the experimental results obtained in the present work, with a general good agreement between experimental results and simulations. The exception is the failure in the mechanism to predict the experimentally early CO formation found, for all the conditions studied, and from experiments carried out in different experimental installations. Additionally, the mechanism has been used to identify the main reactions governing the conversion of CH₃SH and the reactions to which the results are more sensitive.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2019.01.017.

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